COSC 39: Theory of Computation

Problem 1. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

TRIANGLE

- **Input:** An undirected graph G
- **Output:** Does graph G contain a triangle?

Solution. TRIANGLE is in P because given a graph $G = (V, E)$ we can check if every triple of vertices (u, v, w) and test whether all three edges (u, v), (v, w), and (u, w) are present. This requires $|V|C_3$ look ups which is $O(|V|^3)$.

Problem 2. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

MAXCLIQUE

- **Input:** An undirected graph G, a parameter k
- **Output:** Does graph G contain a clique of size at least k?

Solution. MAXCLIQUE is in NP since given a set of vertices we can check if each pair is connected by an edge in polynomial time.

Problem 3. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

PRIMEPOWER

- **Input:** An integer n, and a prime p
- **Output:** Is $n = p^k$ for some positive integer k?

Solution. PRIMEPOWER is in P because we can loop over k until $p^k \geq n$ with $O\left(\left\lceil \log_p(n) \right\rceil\right)$.

Problem 4. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

COMPOSITES

- **Input:** An integer n
- **Output:** Is $n = pq$ for some positive integers $p, q > 1$?

Solution. COMPOSITES is in NP, because if someone else provided integers p, q to us we can verify if $pq = n$ in polynomial time.